

WILLIAMS 35 NNW, AZ, USA

WMO: 722770

Lat: 35.7552N

Lon: 112.3374W

Elev: 1826

StdP: 81.23

Time Zone: -7.00 (W07)

Period: 08-19

WBAN: 53155

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-11.8	-9.2	-22.2	0.6	5.1	-19.6	0.8	4.8	9.4	2.3	8.4	2.5	1.4	N/A	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	32.9	13.2	31.7	13.2	30.5	13.1	16.9	24.4	16.4	24.3	15.8	23.9	4.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.9	13.2	18.9	14.0	12.5	18.7	13.3	11.9	18.7	54.8	24.4	53.0	24.0	51.1	23.9	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.4	8.1	7.0	DB	-17.6	35.2	3.2	1.1	-19.9	36.0	-21.7	36.6	-23.5	37.2	-25.9	38.0	
			WB	-17.7	18.1	3.2	0.5	-20.0	18.5	-21.9	18.8	-23.7	19.1	-26.0	19.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	1.5	2.7	6.7	9.5	13.3	20.2	22.3	21.1	17.9	11.7	6.2	0.9	(6)
(7)		DBStd	8.34	4.24	4.79	3.68	3.94	3.98	2.98	1.80	1.91	2.67	3.74	4.18	4.72	(7)
(8)		HDD10.0	1083	265	204	111	55	17	0	0	0	1	27	122	282	(8)
(9)		HDD18.3	2907	523	437	361	265	158	16	0	3	37	204	363	541	(9)
(10)		CDD10.0	1531	0	1	9	40	121	304	383	344	239	81	9	0	(10)
(11)		CDD18.3	313	0	0	0	0	3	71	124	89	25	1	0	0	(11)
(12)		CDH23.3	4571	0	0	0	16	183	1426	1396	1058	463	29	0	0	(12)
(13)		CDH26.7	1690	0	0	0	1	26	612	577	376	97	1	0	0	(13)
(14)	Wind	WSAvg	2.6	2.3	2.6	3.0	3.4	3.1	3.0	2.4	2.2	2.3	2.4	2.5	2.5	(14)
(15)	Precipitation	PrecAvg	345	29	29	25	20	11	10	46	55	36	31	18	30	(15)
(16)		PrecMax	527	117	93	53	44	44	51	86	115	91	109	73	74	(16)
(17)		PrecMin	195	0	1	0	1	1	0	0	8	2	0	0	2	(17)
(18)		PrecStd	97	30	21	17	13	12	12	22	27	25	35	17	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	19.1	22.2	25.2	29.1	34.9	34.2	32.9	30.9	26.1	21.7	15.8	(19)
(20)			MCWB	4.5	6.0	7.9	9.1	10.4	13.3	14.3	13.4	12.9	10.5	8.0	5.2	(20)
(21)		2%	DB	12.8	16.9	19.9	23.2	27.2	33.4	32.6	31.6	29.1	24.0	19.3	13.3	(21)
(22)			MCWB	3.7	5.4	6.8	8.3	10.0	12.7	14.0	13.9	12.8	9.1	6.6	4.0	(22)
(23)		5%	DB	11.0	14.2	17.9	21.4	25.5	31.9	31.5	30.4	27.7	22.5	17.2	11.1	(23)
(24)			MCWB	3.0	4.7	6.0	7.5	9.1	12.1	14.0	14.1	12.4	9.1	6.0	3.2	(24)
(25)		10%	DB	8.6	11.2	15.8	19.5	23.7	30.4	30.3	29.0	26.2	20.7	15.1	8.5	(25)
(26)			MCWB	2.0	3.4	5.2	6.7	8.7	11.7	13.8	14.2	12.3	8.7	5.4	2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	7.8	8.4	10.1	11.3	15.1	17.8	17.6	16.2	13.1	8.8	6.9	(27)
(28)			MCDWB	11.3	14.7	20.7	23.4	25.7	28.2	24.9	24.6	23.3	19.5	18.9	11.5	(28)
(29)		2%	WB	5.0	6.1	7.4	9.0	10.5	14.1	17.1	16.9	15.4	11.7	7.6	5.7	(29)
(30)			MCDWB	10.4	14.8	18.8	21.7	24.7	29.3	24.5	24.3	22.9	19.2	16.8	10.6	(30)
(31)		5%	WB	3.9	5.2	6.6	8.0	9.7	13.4	16.6	16.3	14.7	10.7	6.6	4.5	(31)
(32)			MCDWB	9.0	13.2	17.0	20.1	23.6	28.9	24.5	24.1	22.6	18.9	15.2	9.3	(32)
(33)		10%	WB	2.9	4.2	5.6	7.2	8.9	12.5	16.0	15.8	13.9	9.7	5.7	3.2	(33)
(34)			MCDWB	8.3	11.3	15.0	18.8	22.2	28.1	24.1	24.0	22.1	17.9	13.7	8.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	12.8	14.9	15.8	16.1	18.4	15.0	14.9	15.4	15.2	14.3	11.4	(35)
(36)			MCDBR	15.2	17.7	18.1	18.9	18.7	20.0	17.4	17.4	17.5	17.7	17.2	15.3	(36)
(37)			MCWBR	8.8	9.4	8.6	8.8	8.2	7.9	5.8	5.8	6.8	8.0	8.8	8.6	(37)
(38)		5% WB	MCDWB	12.9	15.9	16.6	17.3	17.6	17.7	12.7	13.1	13.6	13.5	15.4	12.5	(38)
(39)			MCWBR	7.6	8.6	8.1	8.3	7.8	6.9	3.9	4.7	5.6	6.2	8.1	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.241	0.246	0.262	0.278	0.291	0.295	0.344	0.351	0.297	0.278	0.246	0.239	(40)	
(41)		taud	2.635	2.631	2.577	2.517	2.530	2.504	2.446	2.414	2.576	2.581	2.633	2.643	(41)	
(42)		Ebn at Noon	990	1017	1018	1007	990	980	931	920	965	963	974	972	(42)	
(43)		Edh at Noon	70	80	92	104	104	107	112	114	92	83	70	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.10	4.00	5.52	6.91	7.72	8.34	6.78	6.29	5.87	4.62	3.50	2.74	(44)	
(45)		RadStd	0.27	0.37	0.38	0.35	0.57	0.45	0.47	0.37	0.26	0.36	0.16	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74	+26

Nomenclature: See separate page